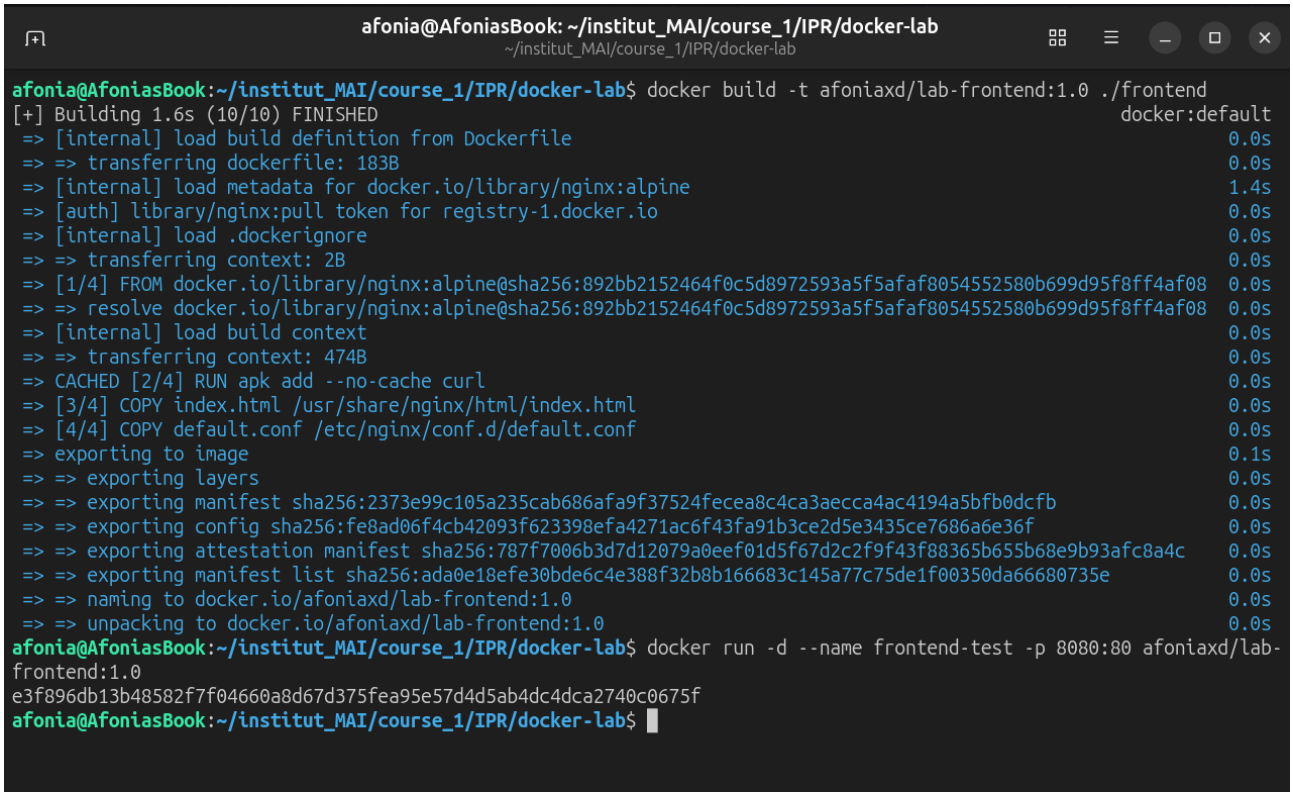
1.1-1.2 Сборка и запуск образа





Лабораторная по Docker

ФИО: Афонин Артём Евгеньевич

GitHub: AfoNiaXd

Docker Hub: afoniaxd

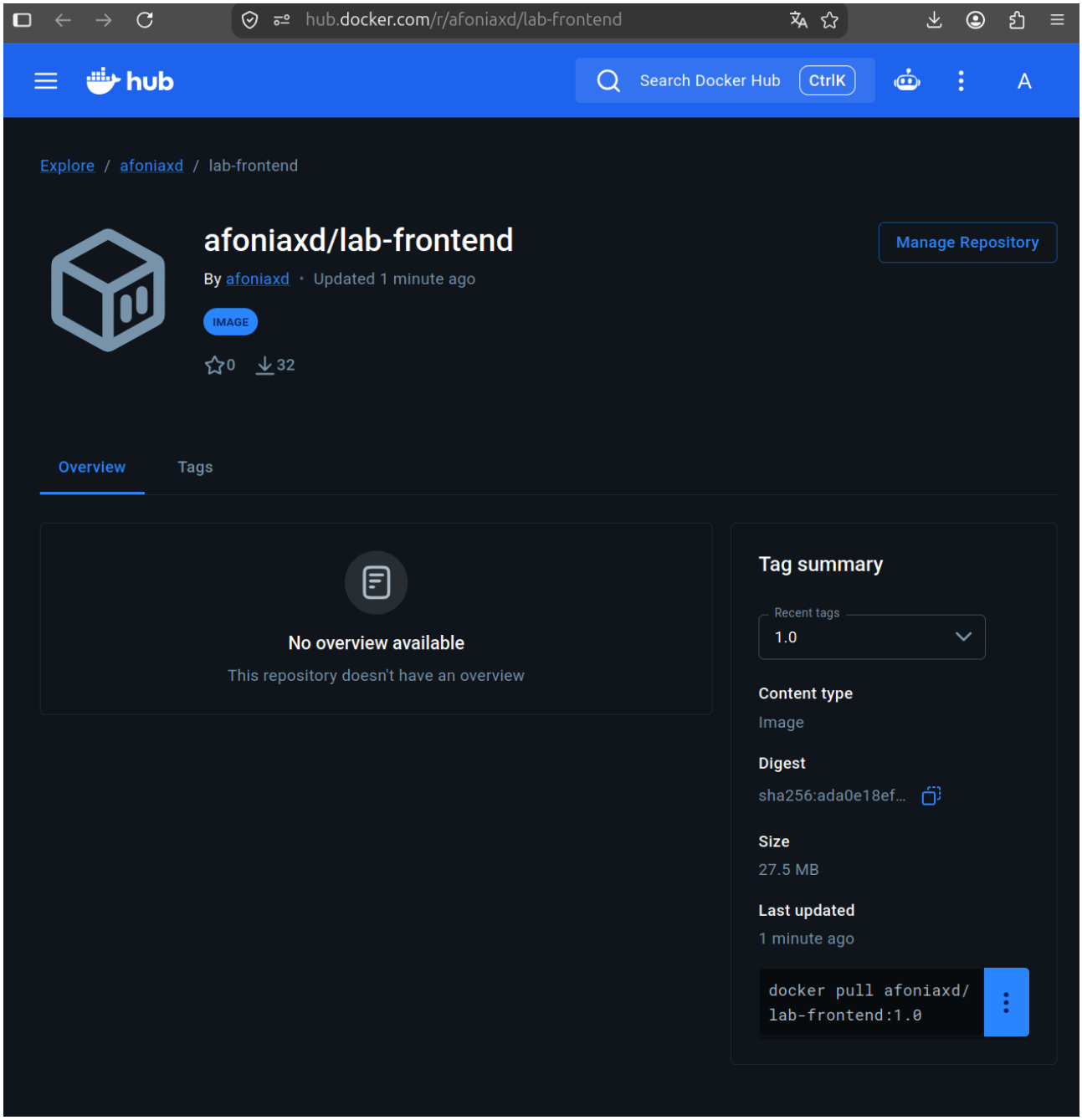
1.3 Push на Docker Hub

```
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker login
Authenticating with existing credentials... [Username: afoniaxd]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker push afoniaxd/lab-frontend:1.0
The push refers to repository [docker.io/afoniaxd/lab-frontend]
e2de96513ba9: Layer already exists
2c11656774f8: Pushed
44136fa355b3: Already exists
6711e0c4f5a7: Layer already exists
b44956913298: Layer already exists
3bf3671a4e6a: Layer already exists
710ec83ca6cd: Layer already exists
70683d58fa5c: Layer already exists
6638b436592c: Layer already exists
1fdd0614a47e: Layer already exists
8723ae51aadd: Layer already exists
90722839de86: Pushed
4a49b27aae0b: Pushed
1.0: digest: sha256:ada0e18efe30bde6c4e388f32b8b166683c145a77c75de1f00350da66680735e size: 856
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$
```

1.4 Образ на Docker Hub



1.5 Удаление и повторный pull

```
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker rm -f frontend-test
frontend-test
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker rmi afoniaxd/lab-frontend:1.0
Untagged: afoniaxd/lab-frontend:1.0
Deleted: sha256:ada0e18efe30bde6c4e388f32b8b166683c145a77c75de1f00350da66680735e
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker pull afoniaxd/lab-frontend:1.0
1.0: Pulling from afoniaxd/lab-frontend
4a49b27aae0b: Pull complete
90722839de86: Pull complete
8723ae51aadd: Pull complete
44136fa355b3: Download complete
2c11656774f8: Download complete
Digest: sha256:ada0e18efe30bde6c4e388f32b8b166683c145a77c75de1f00350da66680735e
Status: Downloaded newer image for afoniaxd/lab-frontend:1.0
docker.io/afoniaxd/lab-frontend:1.0
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$
```

2.2 Запуск и статус сервисов

```
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker compose up -d
[+] up 6/6
 ✔ Network docker-lab_default      Created                                                    0.4s
 ✔ Network docker-lab_appnet       Created                                                    0.7s
 ✔ Volume docker-lab_logger_data   Created                                                    0.0s
 ✔ Container docker-lab-logger-1   Started                                                    0.9s
 ✔ Container docker-lab-api-1      Started                                                    1.0s
 ✔ Container docker-lab-frontend-1 Started                                                    1.6s
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker compose ps
NAME                   IMAGE                       COMMAND                  SERVICE    CREATED              STATUS
          PORTS
docker-lab-api-1       containous/whoami           "/whoami"                api        About a minute ago   Up About
a minute   80/tcp
docker-lab-frontend-1  afoniaxd/lab-frontend:1.0   "/docker-entrypoint.…"   frontend   About a minute ago   Up About
a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
docker-lab-logger-1    busybox                     "sh -c 'while true; …"   logger     About a minute ago   Up About
a minute
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$
```

2.3 Проверка DNS по имени сервиса

```
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker compose exec frontend sh
/ # curl http://api
Hostname: 09eb62883eb5
IP: 127.0.0.1
IP: ::1
IP: 172.20.0.2
RemoteAddr: 172.20.0.3:56218
GET / HTTP/1.1
Host: api
User-Agent: curl/8.22.0
Accept: */*

/ # exit
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$
```

2.5 Логи логгера

```
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker compose exec logger cat /data/log.txt
Mon Oct 19 14:11:07 UTC 2026 - log entry
Mon Oct 19 14:11:12 UTC 2026 - log entry
Mon Oct 19 14:11:17 UTC 2026 - log entry
Mon Oct 19 14:11:22 UTC 2026 - log entry
Mon Oct 19 14:11:27 UTC 2026 - log entry
Mon Oct 19 14:11:32 UTC 2026 - log entry
Mon Oct 19 14:11:37 UTC 2026 - log entry
Mon Oct 19 14:11:42 UTC 2026 - log entry
Mon Oct 19 14:11:47 UTC 2026 - log entry
Mon Oct 19 14:11:52 UTC 2026 - log entry
Mon Oct 19 14:11:57 UTC 2026 - log entry
Mon Oct 19 14:12:02 UTC 2026 - log entry
Mon Oct 19 14:12:07 UTC 2026 - log entry
Mon Oct 19 14:12:12 UTC 2026 - log entry
Mon Oct 19 14:12:17 UTC 2026 - log entry
Mon Oct 19 14:12:22 UTC 2026 - log entry
Mon Oct 19 14:12:27 UTC 2026 - log entry
Mon Oct 19 14:12:32 UTC 2026 - log entry
Mon Oct 19 14:12:38 UTC 2026 - log entry
Mon Oct 19 14:12:43 UTC 2026 - log entry
Mon Oct 19 14:12:48 UTC 2026 - log entry
Mon Oct 19 14:12:53 UTC 2026 - log entry
Mon Oct 19 14:12:58 UTC 2026 - log entry
Mon Oct 19 14:13:03 UTC 2026 - log entry
Mon Oct 19 14:13:08 UTC 2026 - log entry
Mon Oct 19 14:13:13 UTC 2026 - log entry
Mon Oct 19 14:13:18 UTC 2026 - log entry
Mon Oct 19 14:13:23 UTC 2026 - log entry
```

3.3 down -v удаляет том

```
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker compose down -v
[+] down 6/6
 ✔ Container docker-lab-frontend-1 Removed                                    0.4s
 ✔ Container docker-lab-logger-1   Removed                                   10.2s
 ✔ Container docker-lab-api-1      Removed                                    0.2s
 ✔ Volume docker-lab_logger_data   Removed                                    0.0s
 ✔ Network docker-lab_default      Removed                                    0.4s
 ✔ Network docker-lab_appnet       Removed                                    0.7s
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker compose up -d
[+] up 6/6
 ✔ Network docker-lab_appnet       Created                                    0.4s
 ✔ Volume docker-lab_logger_data   Created                                    0.0s
 ✔ Network docker-lab_default      Created                                    0.7s
 ✔ Container docker-lab-api-1      Started                                    1.4s
 ✔ Container docker-lab-frontend-1 Started                                    2.0s
 ✔ Container docker-lab-logger-1   Started                                    0.5s
afonia@AfoniasBook:~/institut_MAI/course_1/IPR/docker-lab$ docker compose exec logger cat /data/log.txt
Mon Oct 19 14:29:51 UTC 2026 - log entry
Mon Oct 19 14:29:56 UTC 2026 - log entry
Mon Oct 19 14:30:01 UTC 2026 - log entry
Mon Oct 19 14:30:06 UTC 2026 - log entry
Mon Oct 19 14:30:11 UTC 2026 - log entry
Mon Oct 19 14:30:16 UTC 2026 - log entry
Mon Oct 19 14:30:21 UTC 2026 - log entry
Mon Oct 19 14:30:26 UTC 2026 - log entry
Mon Oct 19 14:30:31 UTC 2026 - log entry
Mon Oct 19 14:30:36 UTC 2026 - log entry
Mon Oct 19 14:30:41 UTC 2026 - log entry
Mon Oct 19 14:30:46 UTC 2026 - log entry
```